Advaith Anand

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EDUCATION

University of California, Berkeley *Expected May 2029*
B.S. Electrical Engineering & Computer Sciences and Business Administration — M.E.T. 4.0/4.0 GPA
Fall 2026 Coursework: The Structure and Interpretation of Computer Programs (CS 61A), Introduction to Signals, Dynamics, and Information (EE 66), Foundations of Business (UGBA 10X), Managing Technological Innovation (UGBA 196)

EXPERIENCE

Valley Consulting Group — Technical Consultant *Sep 2026 – Present*

- Developing a **computer vision pipeline** for **Packaging Matters** to analyze manufacturing changeover footage, measure process durations, and identify production bottlenecks.

UCLA Visual Machines Group — Student Researcher *Jul 2024 – Jul 2025*

- Achieved **4.8 bpm mean absolute error** in heart-rate estimation across **50+ subjects** by implementing rPPG pipelines, preprocessing facial video, and comparing deep-learning predictions with measurements on UBFC-rPPG.
- Identified **15% higher heart-rate estimation error** among darker-skin-tone subjects by comparing demographic performance and prediction uncertainty, informing recommendations for representative training data and evaluation.

UCLA COSMOS — Student Researcher *Jul 2024 – Aug 2024*

- Reduced word error rate from **28% to 21%** on held-out children's speech by fine-tuning OpenAI Whisper on **500+ audio samples from 40+ speakers**, analyzing **1,000+ transcription errors** to identify recognition failures.

LEADERSHIP

theCoderSchool — Coding Instructor *Aug 2025 – May 2026*

- Taught Python and Scratch to **10+ students**, adapting lessons to individual skill levels and learning pace.

Monta Vista Competitive Programming Club — President *2022 – 2026*

- Supported **80% of participants in advancing to USACO Silver and 40% to Gold** by leading training for **30+ students** in graph algorithms, greedy methods, and dynamic programming through guided problem-solving sessions.

Monta Vista App & Web Development Club — Vice President *2022 – 2026*

- Led student teams developing web and iOS applications by teaching HTML/CSS, JavaScript, and Swift, guiding Git/GitHub workflows, and mentoring members through project planning, implementation, and debugging in Xcode.

PROJECTS

Scale-to-Zero LLM Inference Engine

- Served **16 concurrent requests at 850 ms p95 time to first token** on a benchmark workload by building a Kubernetes inference stack with vLLM, containerized serving, and request routing across **3 LLM sizes**.
- Autoscaled inference capacity from **0 to 4 replicas**, achieving **95-second cold-start readiness** by integrating Prometheus queue-depth metrics with KEDA and scaling serving pods to zero during idle periods.

AI Document Review Platform

- - Reduced average document review time by **40% across 100 test documents** by automating required-field, formatting, and cross-field consistency checks for contracts, applications, invoices, and intake forms.
- Achieved **92% precision in flagging validation issues** against manually labeled examples by detecting missing signatures, conflicting names or dates, and incorrect invoice totals, with field-level explanations and revision suggestions.

Parkinson's Breathing Analysis & Symptom Tracker

- Achieved **0.86 AUROC** in classifying Parkinson's versus healthy controls from **200+ respiratory samples** by extracting breathing-rate and variability features and training a Random Forest with subject-separated cross-validation.
- Enabled **speech- and caregiver-based symptom logging across 8 categories** by building a patient-facing tracker with speech-to-text and alternative entry workflows designed for users with limited fine-motor control.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, Swift, SQL, HTML/CSS
Frameworks & ML/Data: React, Flask, PyTorch, TensorFlow, scikit-learn, pandas, NumPy
Infrastructure & Tools: Kubernetes, vLLM, KEDA, Prometheus, Linux, Git/GitHub, Xcode

HONORS

USA Computing Olympiad Gold Division — Advanced to Gold; one of **4,656 qualifiers**.
Monta Vista Outstanding Student — Selected as **1 of 2 students from 404** for community impact.
Stanford ASES Pitch Competition — Earned **2nd place** in a technology startup pitch competition.